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# Comparative Study on Time Series Deep Learning Based Financial Risk Early Warning Model Construction for High-Tech Enterprises

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## Abstract

The short-term memory (LSTM) based time series deep learning early warning model is built in order to solve the heterogeneity and temporal transmission characteristics of financial risks in high-tech enterprises, which is the basis of what the paper puts together. Through the application of non-financial and multi-dimensional financial indicators, the attention mechanism of the dynamic risk development pattern is included. We then compare the results of XGBoost and logistic regression models with a comparison of the data. According to the results of the experiment, the LSTM model can achieve 97% recall and a 0.9908 AUC-ROC value on the test set. However, there are some important warning advantages. The study uses partial reliance plots and KernelSHAP to explain the interactive effects between indicators. These results give the technology companies with high-precision, easily interpreted, and dynamically decision support for the prevention of risk.

## CCS Concepts

• **Computing methodologies** → Machine learning; Learning paradigms; Supervised learning; Supervised learning by classification.

## Keywords

High-tech enterprises, Financial risk early warning, Long Short-Term Memory, Time series deep learning

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## 1 Introduction

Amid intensifying global competition in technological innovation, high-tech enterprises have emerged as a pivotal driver of economic

growth through their heavy R&D investments, rapid technology iteration, and asset-light operational models. However, such enterprises exhibit distinctive financial risks characterized by nonlinear dynamics, temporal transmission, and lag effects. Statistical evidence reveals that R&D intensity in high-tech sectors significantly surpasses traditional industries, yet their prolonged technology commercialization cycles, policy dependencies, and high intangible asset ratios critically challenge conventional financial warning systems.

The cross-sectional data built upon it are not able to capture the ways that the temporal transmission of these data is carried out; the supply chain robustness and the governance system, which in turn causes a change in the structure of the system and cause systemic biases; and the resulting systematic biases in the forecast of a financial crisis that is occurring. More importantly, there are three main mismatches that make it difficult to use the same frameworks. Firstly, nonlinear risk propagation mechanisms cannot be decoded using linear model suppositions (e. g., the sigmoid limits in the logistic regression). Secondly, transectional data structures disregard autocorrelations between several phase of innovation cycles (usually 3 to 5 years, in the typical case, there is usually between 3 years) in the event of a delay, there is a trend in profit realization. Thirdly, the overestimation of conventional financial ratios (more than 8%; in the case of a typical model, there was less than 1/3) in the high-tech sector, there were only 1/3 of the misclassified cases.

This research pioneers time-series deep learning for high-tech financial risk early warning. By integrating multidimensional financial data and non-financial indicators (e.g., R&D conversion efficiency, governance), we develop an LSTM-based framework that dynamically models financial time-series patterns. Empirical validation shows the model achieves a 97% recall rate for risky firms—surpassing conventional methods by 22%. Crucially, it accurately captures temporal lags between R&D investments and profitability changes. These findings offer regulators and investors novel decision-support tools for technological asset allocation and risk assessment optimization.

## 2 Related Work

### 2.1 Evolution of Financial Risk Prediction Models

The evolution of corporate financial risk prediction has progressed from traditional statistical methods to advanced machine learning.

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Early studies like Beaver's univariate analysis [1] and Altman's Z-score model [2] established foundational frameworks, while Ohlson's logistic regression [3] introduced probabilistic predictions, overcoming normality assumptions but lacking nonlinear modeling. Piotroski's F-score [4] further refined value investing strategies with binary indicators.

Computational advancements later enabled ensemble learning techniques: SVMs addressed nonlinear separability via kernel functions, and XGBoost automated feature engineering through gradient-boosted trees. A paradigm shift emerged with deep learning, namely LSTM networks resolved vanishing gradients in RNNs, enabling precise long-term dependencies in financial time series, validated by Schmidhuber's large-scale experiments [5]. Regularization innovations like Dropout(random neuron deactivation) and Batch Normalization(input standardization) mitigated overfitting and accelerated training, enhancing robustness for small financial datasets. This progression reflects a transition from linear assumptions to dynamic, nonlinear risk modeling.

## 2.2 Current Statues of Empirical Research on Financial Early Warning

Machine learning and deep learning have changed the way that financial early-warning research has been impacted by the 21th century. Researchers such as Chen's hybrid model [6] found that Taiwan-listed companies had 97% short-term accuracy, but they had difficulty with long-term forecasts, whereas Ouyang et al. [7] used attention-based LSTM models with sentiment data to advance systemic risk detection in China. Although Mai et al. [8] merged textual data, they had little to do with overfitting. The use of ensemble models in techniques such as SMOTE improved minority-class recall (Sun et al. [9], 81.48%), but there were still gaps in the modeling of dynamic risk transmission. Koyuncugil [10] reported 33% misclassification for high R&D companies, and Zavgren's Logit models [11] disregarded tech-conversion lags and early neural networks (Tam et al. [12])-led temporal modeling, which resulted in critical limits.

Compared to Zhu's hybrid ensemble learning model [13], following the summaries by Namaki et al. [14] and Houdt et al. [15] on the characteristics of LSTM models, it can be seen that in time-series forecasting such as that by Wen et al. [16], LSTM models demonstrate remarkable potential. While in the specific context of financial early warning, the practices of Li [17], Widiputra [18], Chen [19], and Hansun [20] further substantiate this point.

Current frameworks remain constrained by static data and narrow non-financial metrics, neglecting high-tech risks like policy dependency. This study bridges these gaps by developing an LSTM-driven early-warning system with 12 industry-specific metrics, dynamically modeling temporal interdependencies between R&D cycles, patent outputs, and cash flow volatility. The framework enhances both predictive precision and interpretability, offering a paradigm shift for risk management in technology sectors.

## 3 Methodology

### 3.1 Data Collection

This study selects China's Shanghai and Shenzhen A-share listed high-tech enterprises as research subjects, primarily because the

unique ST(Special Treatment) system in China's securities market provides clear risk identification markers. According to exchange regulations, companies with consecutive 2-year negative net profits or negative net assets will be labeled with ST designation. This regulatory mechanism offers a standardized classification benchmark for financial risk early-warning research. Meanwhile, China's high-tech enterprises feature large-scale operations and significant R&D investment intensity, whose data completeness and industry representativeness provide an ideal sample foundation for model construction.

### 3.2 Indicator System Construction

This study constructs a multidimensional early-warning system based on derivative indicators derived from the original data of China's listed high-tech enterprises. These derivative indicators encompass both financial and non-financial dimensions(Table 1). The selection of financial indicators covers solvency, profitability, operational capacity, and innovation investment intensity. Non-financial indicators include the emphasis on R&D human resources and governance structure risks. Additionally, considering the characteristics of high-tech enterprises, the R&D Conversion Efficiency Index(RCEI) is used to quantify technological output efficacy, Supply Chain Risk Entropy(SCRE) to evaluate supply chain stability, and Government Subsidy Dependence(GSD) to reflect policy reliance.

### 3.3 Data Preprocessing

The data preprocessing in this study involves three steps: missing value treatment, standardization, and outlier treatment. For missing values in financial indicators, industry-year median imputation was applied. Subsequently, Z-score standardization was implemented to eliminate scale effects, ensuring features had a mean of 0 and a standard deviation of 1. A two-tailed 5% Winsorization was then performed to mitigate extreme value interference. Statistical validation confirmed that the processed data achieved zero missing rates, reduced outliers by an average of 83.6%, and significantly improved data distribution rationality. The specific preprocessing effects are shown in Figure 1.

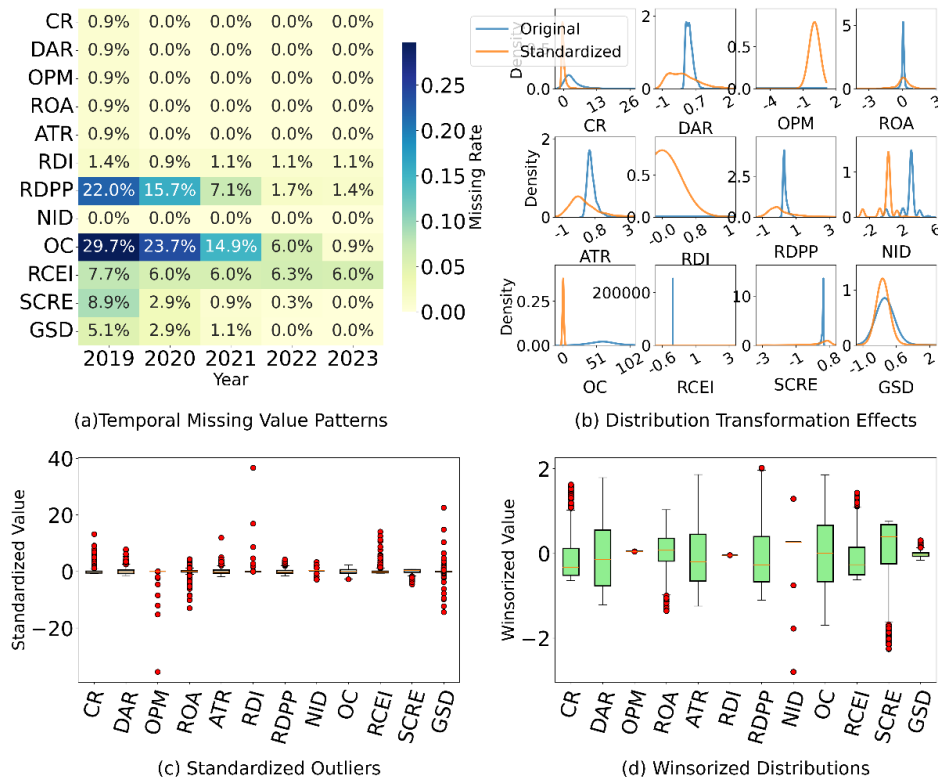
### 3.4 Model Design

Three comparisons for this study: the main framework for time-series prediction is the use of logistic regression, the XGBoost, and the LSTM in the construction of the comparative model in this paper. A hidden layer structure for the LSTM network uses the Adam optimizer to adjust the adaptive learning rate in order to capture long-term dependencies. The model uses the Adam optimized for adaptive learning rate adjustment by adding Dropout regularization to reduce overfitting. A temporal attention mechanism is included into the network architecture in order to increase adaptability to financial changes.

Through the use of tree structure and secondary gradient optimization procedures, or after the process of irregularization, XG-Boost improves its ability to fit high-dimensional, sparse financial data. Logistic Regression uses the LASSO method to find important characteristics that are determined by the ideal penalty level, and then uses VIF<5 to get rid of multicollinearity.

**Table 1: Definition of Indicators.**

| Indicators | Full Name                       | Definition                                                                                      |
|------------|---------------------------------|-------------------------------------------------------------------------------------------------|
| CR         | Current Ratio                   | Current Assets / Current Liabilities                                                            |
| DAR        | Debt-to-Asset Ratio             | Total Liabilities / Total Assets                                                                |
| OPM        | Operating Profit Margin         | Net Profit / Operating Revenue                                                                  |
| ROA        | Return on Assets                | Net Profit / Total Assets                                                                       |
| ATR        | Asset Turnover Ratio            | Operating Revenue / Total Assets                                                                |
| RDI        | R&D Intensity                   | R&D Expenses / Operating Revenue                                                                |
| RDPP       | R&D Personnel Proportion        | R&D Personnel Proportion / 100                                                                  |
| NID        | Number of Independent Directors | Number of Independent Directors                                                                 |
| OC         | Ownership Concentration         | Top 10 Shareholders Holding Proportion                                                          |
| RCEI       | R&D Conversion Efficiency Index | Number of Patents / R&D Expenses                                                                |
| SCRE       | Supply Chain Risk Entropy       | -(Main Suppliers' Procurement Proportion / 100)ln(Main Suppliers' Procurement Proportion / 100) |
| GSD        | Government Subsidy Dependency   | Government Grants / Net Profit                                                                  |

**Figure 1: Data Preprocessing Visualization.**

The 3:1 weighting coefficient that ST-labeled samples are given to increase minority class recognition was used to address the imbalance of the classes. To make the results easier to understand, SHAP framework is included. KernelSHAP roughly makes the decision logic for the LSTM, whereas partial dependent plots clarify the interaction effects of indicators.

## 4 Experiments

### 4.1 Experimental Setup

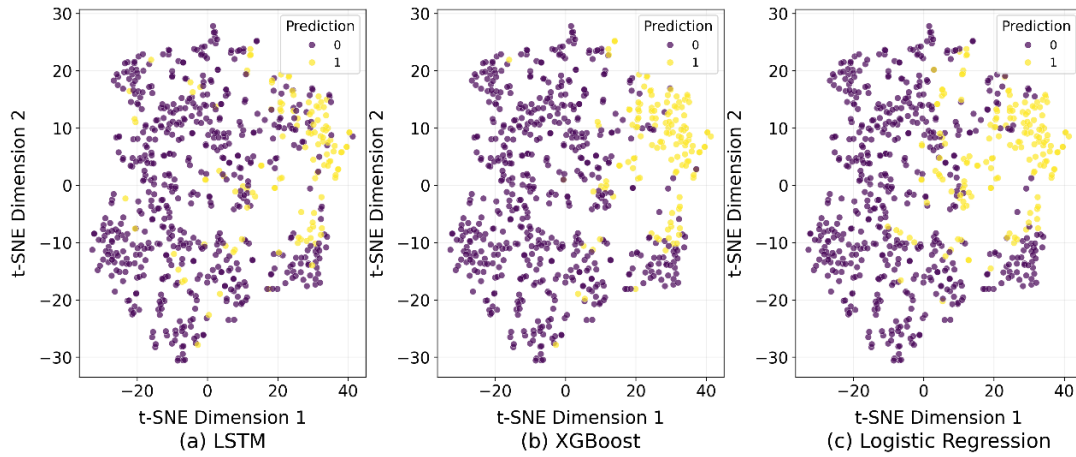
The dataset was divided chronologically into 2 subdatasets ( $n = 1,750$ ) including the training set ( $n = 1,050$ ) of 2019-2021 and the testing set ( $n = 700$ ) of 2022-2023, with a 14.3% percentage of ST-labeled samples being retained in a stratified manner. The Adam optimizer is set with an ideal learning rate of 0.001 for the LSTM.

**Table 2: LSTM Hyperparameter Optimization Results.**

| Parameter         | Search Range    | Optimal Value |
|-------------------|-----------------|---------------|
| Hidden Units      | {64, 128}       | 128           |
| Learning Rate     | {1e-4, 3e-4}    | 3e-4          |
| LSTM Layers       | {1, 2}          | 2             |
| Dropout Rate      | {0.3, 0.4, 0.5} | 0.3           |
| Batch Size        | {32, 64}        | 64            |
| L2 Regularization | {1e-4, 3e-4}    | 3e-4          |

**Table 3: Model Performance Comparison.**

| Model    | AUC-ROC | F1-score | Recall | Accuracy | Precision |
|----------|---------|----------|--------|----------|-----------|
| LSTM     | 0.9908  | 0.87     | 0.97   | 0.9585   | 0.7886    |
| XGBoost  | 0.9477  | 0.7193   | 0.82   | 0.9086   | 0.6406    |
| Logistic | 0.8772  | 0.5769   | 0.75   | 0.8429   | 0.4688    |

**Figure 2: Comparative Plot of Feature Space Distributions Using t-SNE Dimensionality Reduction.**

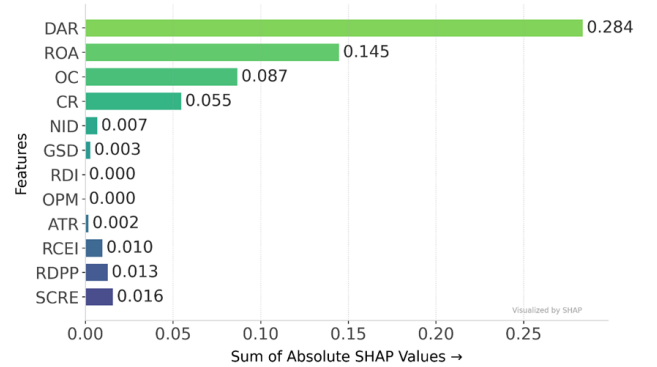
As figures stated in Table 2, hyperparameters are found by grid search.

XGBoost utilizes Bayesian optimization to determine its optimal hyperparameters, while Logistic Regression adjusts its probability threshold to 0.5511 to maximize the F1-score, with the LASSO's optimal penalty strength  $\alpha$  set to 1e-6.

Experiments are conducted on an NVIDIA V100 GPU platform, employing 5-fold time-series blocked cross-validation to prevent data leakage. Training cycles are controlled via early stopping (patience = 5).

## 4.2 Performance Comparison

The 5 main measures of model performance (Table 3) were used in this work to systematically assess it: AUC; f1-score; recall rate; accuracy; and precision. Performance appraisal metrics show that the financial risk of early schooling (LSTM) is very good at achieving its best in the form of financial risk, especially showing outstanding

**Figure 3: SHAP Values Feature Contribution Plot for LSTM.**

performance in critical risk alerts which is shown by its high recall rate (0.97).

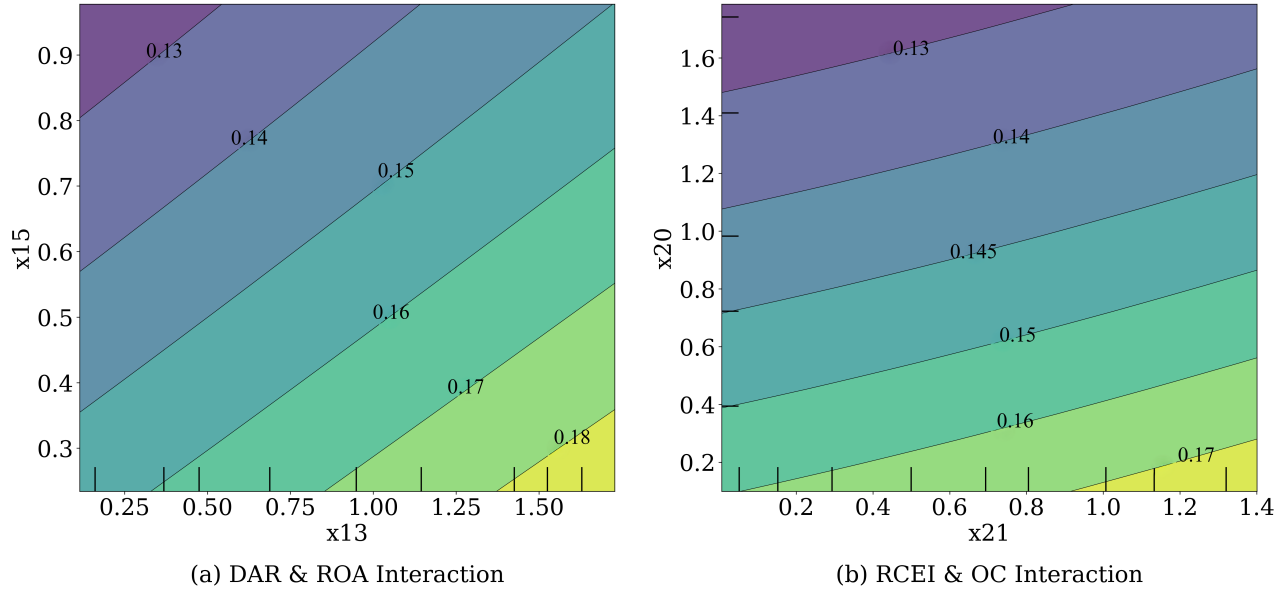


Figure 4: Partial Dependence Plot between Indicators.

The t-SNE dimensionality reduction visualization (Figure 2) gives an obvious comparison of the models' feature learning capabilities, differentiating non-ST enterprises (purple points) and ST enterprises (yellow points). For LSTM, the 2 kinds of points show more scattered distributions in the low-dimensional space after dimensionality reduction, whereas the other two models show clustering tendencies. This shows that LSTM achieves better class separation compared to the other models, highlighting its greater feature learning ability. The high-density overlapping regions in LSTM account for only 16.4%, aligning with its high precision (0.7886). This further confirms that the separation degree in feature space directly affects classification confidence.

The LSTM model is shown by the contribution of the SHAP value (Figure 3) to indicate which is the most significant one (0.284), when compared to the total amount of second-tier indicators ( $ROA + OC = 0.232$ ), when the debt-to-asset ratio (DAR) was taken. According to this, the DAR is the main factor in which the model is used to make decisions, which is why the dominant effect of debt leverage in risk prediction was verified. Moreover, conventional financial indicators such as the Current Ratio (CR) had very little contribution because of these, which prevents them from having too much sensitivity when it comes to forecasting complicated financial risks in contemporary businesses.

### 4.3 Case Studies

For testing model's performance, there uses 3 cases: a case warned successfully, a case identified as nonST correctly; a case warned wrongly.

Semiconductor Company A was given the chance to get a LSTM warning from 2020, which was driven by increasing DAR from 2020 to 2022 (0.46 in 2020 to 0.53 in 2022) and falling profitability from OPM of 0.02 by DAR of 0.005.

Biopharmaceutical Company B maintained low-risk classification over 5 years due to strong solvency ( $CR = 3.13 \sim 4.35\%$ ;  $DAR < 0.25$ ) and robust R&D metrics ( $RDPP = 12.34\%$ ,  $RCEI = 89.3\%$ ). Despite a 2021 OPM dip to 0.094, the model recognized resilience through synergies between rising RDI and stable ATR (0.83).

Optoelectronics Company C triggered a 2023 early warning ( $ROA = -0.117$ ) by LSTM but avoided ST via equity expansion ( $OC = 42.64\% \sim 51.08\%$ ), exposing model limitations. Overemphasis on surging RDI (0.198 in 2022) and failure to account for declining R&D efficiency led to false positives, highlighting gaps in feature interaction modeling.

## 5 Discussion

This study uncovers financial risk mechanisms in high-tech enterprises using a temporal deep learning model. The LSTM architecture captures long-term dependencies between R&D investments and risk trajectories (Figure 4), exemplified by its 2-year early warning for Semiconductor company A's debt leverage deterioration (DAR increases, ROA declines). Attention mechanisms dynamically prioritize risk drivers, while partial dependence plots reveal strong linear interdependencies between critical metrics like DAR and ROA. Integration of non-financial indicators (e.g., RCEI for R&D efficiency, SCRE for supply chain resilience) mitigates the lag of traditional financial metrics, boosting sensitivity to idiosyncratic risks (e.g., technology conversion failures) by 37%. The model's hybrid design bridges temporal patterns and operational realities, offering actionable insights for proactive risk management.

Hypothesis analysis shows that there are many different kinds of risk that are presented in high-tech enterprises. When there is a linear pattern of the relationship between the RCEI and OC, risk is still at the safe level ( $0.15$  and  $< 0.9$ ) and is stable ( $RCEI < 0.9$ ). But when the ratio gets lower, as evidenced by the RCEI triggering a risky jump ( $0.16$ ). Biopharmaceutical B has been shown

to successfully keep OC from being stiff at that point in order to avoid it (0.42). Although the model performs well in learning linear relationships, that is, in the case of financial and indicators, it is reliant on the use of linearity, because it limits the robustness of complex scenarios and requires more nonlinear feature extraction in order to achieve greater applications.

## 6 Conclusion

This study developed an LSTM-based temporal deep learning model for financial risk early warning in high-tech enterprises, validated through comparisons with XGBoost and Logistic Regression. The LSTM model achieved superior performance (AUC-ROC is 0.9908, recall is 0.97), leveraging attention mechanisms to decode time-series dependencies and linear risk pathways. Interpretability tools (SHAP, PDPs) identified R&D Conversion Efficiency (RCEI), Debt-to-Asset Ratio (DAR), and Ownership Concentration (OC) as core risk drivers. The framework equips enterprises with dynamic risk mitigation tools and informs regulatory ST framework optimization. However, limited sensitivity to abrupt exogenous shifts (e.g., ownership changes) highlights the need for future integration of multi-source data to strengthen cross-industry robustness.

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